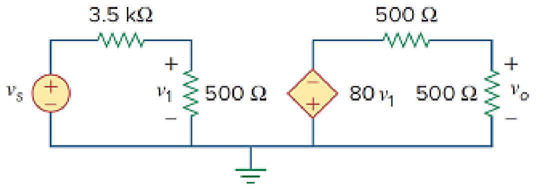
Chapter 3, Problem 87P

Problem

For the circuit in Fig. 3.123, find the gain vo/vs.

Figure 3.123:



Step-by-step solution

Step 1 of 5

Refer to the circuit in Figure 3.123 in the textbook.

Identify the loops and currents in the circuit diagram shown in Figure 1.

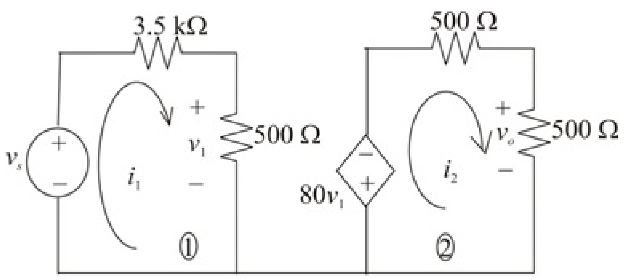


Figure 1

Step 2 of 5

Apply Kirchhoff's Voltage Law in loop-1.

$$-v_s + i_1(3.5\text{ k} + 500) = 0$$
$$v_s = i_1(4\text{ k})$$

$$i_1 = \frac{v_s}{4\text{ k}}$$
$$= (0.25\text{ m})v_s$$

Therefore, the expression for the current i_1 is, $i_1 = (0.25\text{ m})v_s$.

Step 3 of 5

By ohm's law the voltage v_1 is,

$$v_1 = i_1(500)$$

Substitute $(0.25\text{ m})v_s$ for i_1.

$$v_1 = (0.25\text{ m})v_s(500)$$
$$= 0.125v_s$$

Therefore, the expression rot he voltage v_1 is, $v_1 = 0.125v_s$.

Step 4 of 5

Apply Kirchhoff's Voltage Law in loop-2.

$$80v_1 + i_2(500+500) = 0$$
$$80v_1 + i_2(1000) = 0$$

Substitute $(0.125)v_s$ for v_1.

$$80(0.125)v_s + i_2(1000) = 0$$
$$10v_s + 1000i_2 = 0$$
$$1000i_2 = -10v_s$$
$$i_2 = -\left(\frac{10}{1000}\right)v_s$$
$$i_2 = -0.01v_s$$

Therefore, the expression for the current i_2 is, $i_2 = -0.01v_s$.

Step 5 of 5

Consider the expression for the voltage v_o.

$$v_o = i_2(500)$$

Substitute $-0.01v_s$ for i_2.

$$v_o = (-0.01v_s)(500)$$
$$\frac{v_o}{v_s} = -5$$

Therefore, the voltage gain $\frac{v_o}{v_s}$ is, $\boxed{-5}$.